

CryptoPilot

Automated BTC Trading Platform — Business Solution Document

Automated execution of the validated **Trend Rider v6 long/short strategy** on Binance USDT-M Futures — profitable participation in bull markets AND bear markets — with a web dashboard, SMS alerts, pluggable strategies and an AI market-news assistant.

Field	Value
Document	Business Solution Document (BSD)
Version	2.0 — supersedes v1.0 (strategy upgraded v5.2 → v6 long/short)
Date	17 July 2026
Prepared for	Project owner (Binance account holder)
Stack	Python 3.11+ · FastAPI · PostgreSQL 16 · React · Docker
Trading venue	Binance USDT-M Futures — DEMO (testnet) first, then LIVE
Strategy	Trend Rider v6: v5.2 long engine + vol-sized short sleeve (LONGSHORT_REPORT.md)

IMPORTANT — RISK NOTICE: This document describes software that trades real money automatically, including short selling on futures. Backtested performance (2023–2026, fees included, out-of-sample validated) does not guarantee future results. Leveraged futures can lose more than the amount risked per trade. The system enforces layered risk controls (vol-targeted sizing, monthly circuit breakers, leverage caps, kill-switch), but no automated system removes market risk.

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1. Executive summary

CryptoPilot is a self-hosted automated trading platform that executes the owner's validated 4-hour BTCUSDT strategy — **Trend Rider v6** — on Binance, unattended, 24/7. Version 2.0 of this document upgrades the traded strategy from the long-only v5.2 to the long/short v6 composite: the identical long engine plus a volatility-sized, stop-free **short sleeve** that earns during bear markets instead of sitting flat. In the 3-year backtest (real Binance data, fees included) v6 returned +204.5% with a –17.2% maximum drawdown and made **+31.7% in the out-of-sample bear year** in which buy-and-hold lost 41%.

The owner controls the bot from a secure web dashboard: **start / stop at any time** (24/7 loop while running), switch between the Binance **DEMO (testnet)** and **LIVE** accounts in Settings, watch open positions and live profit, and review every historical trade. Portfolio figures (balance, positions, income) are read from the Binance account API — the exchange, not the bot's memory, is the source of truth. Every significant event — trade opened, trade closed, bot started, bot stopped, breaker, error — is pushed to the owner's phone by SMS through **notify.lk**.

The platform is deliberately built in decoupled halves. The trading core is strategy-agnostic: strategies are plugins behind a fixed interface (now including short intents), so v6 can be swapped or A/B-replaced without touching execution, risk, dashboard or notification code. A fully separate **AI news agent** collects daily crypto/macro news, summarises it with the Claude API and presents a morning briefing on the dashboard — informational only, never a trading input.

2. Business goals and success criteria

#	Goal	Success criterion
G1	Trade the validated v6 strategy automatically, both directions	Bot decisions match the reference backtest engine bar-for-bar on the same data (automated parity test)
G2	Earn in bear markets, not just avoid losing	Short sleeve activates in deep-bear regime and its live behaviour matches Appendix A expectations
G3	Safe rollout	≥4 weeks on DEMO with zero unexplained deviations before LIVE is enabled
G4	Owner always in control	Start/Stop/kill-switch work at any moment; SMS arrives ≤60s after each event
G5	Full auditability	Every decision, order, fill, SMS and error reconstructable from PostgreSQL
G6	Swappable strategy	A new strategy class can go live with zero changes outside the strategies package
G7	Informed owner	Daily AI news briefing on dashboard by the configured local time

3. Scope

In scope (v2)

- Automated trading of BTCUSDT USDT-M futures on Binance DEMO (testnet) and LIVE, selectable in Settings.
- Trend Rider v6 exactly as validated: long engine (regime filter, pullback entries, 2.5×ATR stop, TP1 40% at 1R, breakeven, 4×ATR trail, regime exit, 4% monthly breaker) **plus** the short sleeve (deep-bear

regime, 75% × vol-scale sizing, no price stop, cover on regime end, separate 4% monthly sleeve breaker).

- Web dashboard with authentication: bot control, live position, equity curve, trade history, monthly P&L, news briefing, settings.
- SMS notifications via notify.lk; full event log in the database.
- AI news agent with daily scheduled collection and Claude-generated summaries.
- Docker-based deployment on a small always-on VPS.

Out of scope (v2)

- Automatic withdrawals or any transfer of funds (withdrawal permission is never granted to the API key).
- Multi-coin portfolios, grid/HFT strategies, sub-4h timeframes (all tested and rejected — see reports).
- News-driven automated trading decisions (news is informational only).
- Native mobile app (dashboard is responsive; SMS covers push needs).

4. System architecture

CryptoPilot is a modular monolith: one deployable backend with strictly separated internal services plus a React single-page dashboard. Modules communicate through defined interfaces and PostgreSQL — never by reaching into each other's internals — so any module (strategy, notifier, news agent) can be replaced in isolation.

CryptoPilot — Business Solution Diagram

Automated BTCUSD trading of Trend Rider v6 (long + short sleeve) on Binance USDT-M · Python + FastAPI + PostgreSQL · v1.1 · 17 Jul 2026

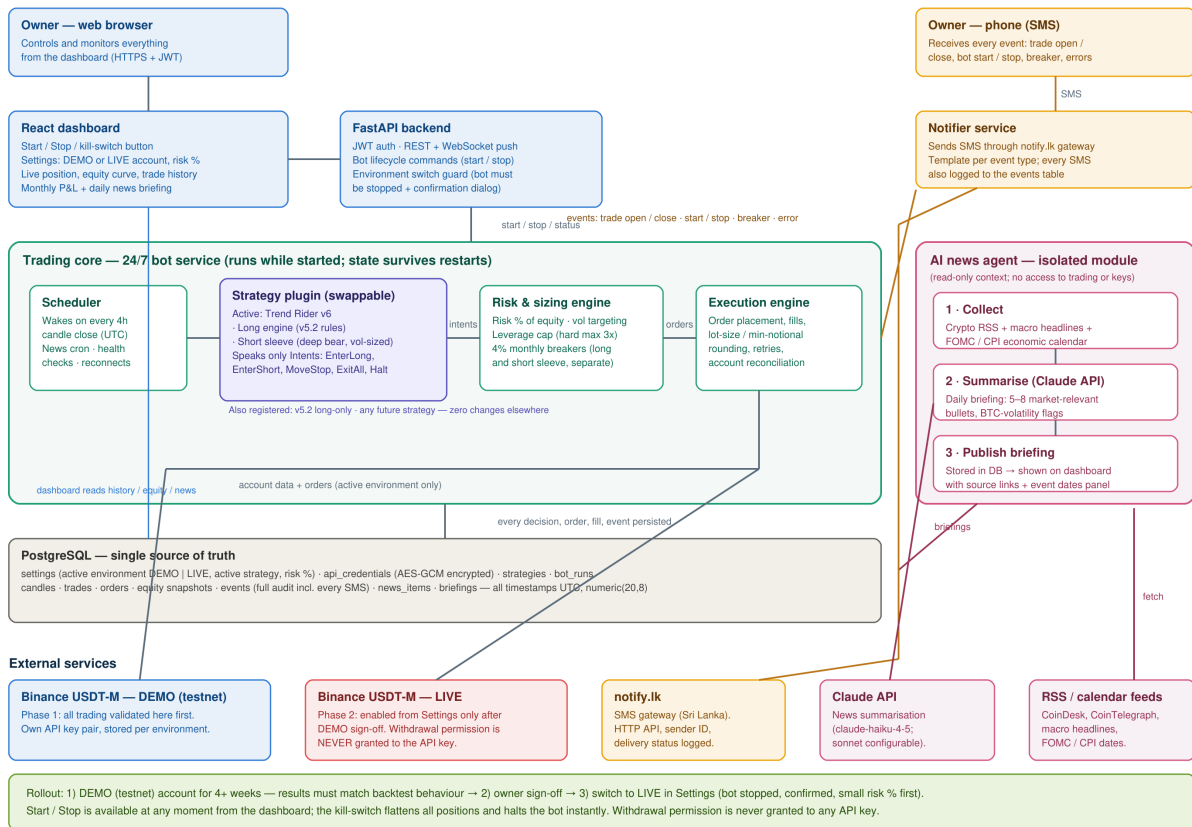


Figure 1 — Component architecture (also delivered standalone: CryptoPilot_Solution_Diagram.pdf).

Component	Responsibility
React dashboard	UI: login, bot Start/Stop/kill-switch, settings (environment, strategy, risk %), live position, equity & trade history, monthly P&L, news briefing.
FastAPI backend	JWT auth, REST API, WebSocket push for live updates, bot lifecycle commands, environment-switch guard (bot must be stopped + confirmation).
Bot service	The 24/7 loop: wakes on every closed 4h candle, feeds the strategy plugin, executes returned intents. State survives restarts (auto-resume).
Strategy plugin	Trend Rider v6 behind the Strategy interface; emits abstract intents only (long AND short). Other strategies registered alongside.
Risk & sizing engine	Risk % of equity, short-sleeve vol targeting, leverage cap, both monthly breakers, exchange minimums.
Execution engine	Order placement/cancel, fills, lot-size rounding, retries, reconciliation against the Binance account.
Scheduler	4h candle-close ticks, news cron, health checks, reconnects.
Notifier	Event → SMS via notify.ik + database event row.

Component	Responsibility
News AI agent	Isolated collect→summarise→publish pipeline (no access to trading or keys).
PostgreSQL	Single source of truth: settings, credentials (encrypted), strategies, runs, trades, orders, equity, events, news.

5. Functional requirements

ID	Requirement	Priority
FR-01	Settings selects the active environment: DEMO (testnet) or LIVE. Each environment stores its own API key pair. Switching requires the bot stopped and shows a confirmation dialog.	Must
FR-02	Dashboard Start/Stop button controls the bot. Start = connect, reconcile account state, begin 24/7 loop. Stop = stop evaluating (position left as-is by default); optional 'Stop & close' flattens safely. State survives server restarts.	Must
FR-03	Bot trades Trend Rider v6 rules exactly as validated (Appendix A), LONG and SHORT, acting only on closed 4h candles.	Must
FR-04	Long sizing: risk % of equity per trade (default 2% live), leverage capped (default 3x). Short sleeve sizing: $\text{weight\%} \times \min(1, \text{vol-target} / \text{realized vol})$ of equity, resize only when target drifts >20%. Quantities rounded to Binance lot size; min-notional respected.	Must
FR-05	Portfolio data (balance, positions, income history) is read from the Binance account API and displayed; the account is the source of truth.	Must
FR-06	SMS via notify.lk for: trade opened (side, price, size), trade closed (P&L, reason), bot started, bot stopped, breaker triggered (long or sleeve), error/pause. Every SMS also logged.	Must
FR-07	Trade history: filterable list of all trades with entry/exit, fees, R-multiple, exit reason, strategy, environment; CSV export.	Must
FR-08	Both monthly circuit breakers enforced engine-side: long book -4% month-to-date → flatten + halt longs till the 1st; short sleeve -4% of equity month-to-date → cover + halt shorts till the 1st (independent).	Must
FR-09	Kill-switch: one click cancels all orders, flattens all positions, stops the bot, sends SMS.	Must
FR-10	AI news agent produces a daily briefing shown on the dashboard with source links and FOMC/CPI calendar panel; on-demand refresh.	Should
FR-11	Strategy selection in Settings from registered plugins with editable parameters (validated defaults pre-filled).	Should
FR-12	Equity snapshots recorded every 4h close; dashboard renders live + historical equity curve vs buy-and-hold.	Should

6. Non-functional requirements

Area	Requirement
Reliability	Bot auto-reconnects (exchange REST/WS), resumes after crash/restart from persisted state, and reconciles against the exchange before acting. Missed-candle catch-up on reconnect.
Latency	Decisions occur once per 4h close; execution within seconds of close is sufficient. No HFT requirements.
Auditability	Every decision, order, fill, SMS and error persisted with UTC timestamps; trades reconcile to Binance income history.
Maintainability	Strategy, notifier and news modules replaceable behind interfaces; typed Python (mypy); strategy math covered by parity tests against the validated backtest engine.

Area	Requirement
Security	Keys encrypted at rest, TLS everywhere, no withdrawal permission, JWT + argon2 + optional TOTP.
Portability	Docker Compose; runs on a \$5–10/month VPS (1 vCPU, 1–2 GB RAM).

7. Strategy plugin framework (loose coupling)

The strategy never talks to Binance and never sizes positions. It receives candles and current trade state and returns abstract **intents**. The execution engine owns sizing, exchange filters and order mechanics. A new idea is a new class, registered by name, selectable in Settings — zero changes to execution, risk, dashboard or notifications. v2.0 extends the intent vocabulary with short-side intents for v6.

Interface element	Meaning
Strategy.name / params	Registered name + parameters editable in Settings (validated defaults).
warmup_bars()	History needed before first decision (v6: 200 bars for SMA200).
on_candle(candles, state) → [Intent]	Called once per closed 4h candle with full account/trade state.
EnterLong(stop_distance, tp_levels)	Open long; engine sizes it from risk % and places stop + TP orders.
EnterShort(weight, vol_target)	v6 sleeve: open short; engine computes vol-scaled notional and applies the sleeve breaker. No price stop by design.
ResizeShort(target_notional)	Adjust short toward vol target when drift >20% (fee-churn guard in engine).
MoveStop(price)	Ratchet-only (engine refuses to lower a long stop).
TakePartial(level_id)	Informational; TP orders rest on the exchange.
ExitAll(reason)	Regime death, breaker, manual, kill-switch.
Halt(until)	Stand aside (monthly breakers).

Registered strategies at launch: **trend_rider_v6** (default), **trend_rider_v52** (long-only fallback). Both are pinned to the exact parameter sets validated in the backtests.

8. Binance integration design (DEMO / LIVE)

- **Environments.** DEMO = Binance USDT-M testnet (testnet.binancefuture.com); LIVE = fapi.binance.com. Same code path; base URL and key pair come from the active environment row. Nothing else differs.
- **Market data.** 4h klines via REST backfill + user-data/kline WebSocket; candles cached in PostgreSQL. Decisions only on candle close (kline 'closed' flag).
- **Orders — long engine.** Entry MARKET on signal close; STOP_MARKET (reduce-only) protective stop; TP1 LIMIT (reduce-only) for 40%; trail managed by the engine via stop modification on each close.
- **Orders — short sleeve.** Entry/resize MARKET (reduce-only on reductions); NO protective price stop (validated design — size is the risk control); cover MARKET on regime end, sleeve breaker or kill-switch.
- **Account.** One-way position mode, isolated margin, leverage set explicitly (cap default 3x, v6 never needs more than ~1x for the sleeve + ~3.4x worst case long at 5% risk).
- **Reconciliation.** On every start and every 4h close the engine compares expected vs actual position/orders; any mismatch → safe-mode (no new entries) + SMS alert.
- **Rate limits & errors.** Exponential backoff, idempotent client order IDs, margin-error (–2019) handling pauses the bot with SMS.

9. Database design (PostgreSQL)

PostgreSQL 16. All timestamps timestamptz (UTC). numeric(20,8) for prices/quantities — never floats for money.

Table	Purpose / key columns
users	Dashboard login: id, email, password_hash (argon2), role, totp_secret, created_at.
app_settings	Singleton: active_environment (DEMO LIVE), active_strategy, risk_pct, sleeve_weight_pct, sleeve_vol_target, leverage_cap, sms_enabled, news_sources[], news_time.
api_credentials	Per environment: environment, api_key, api_secret_encrypted (AES-GCM via master key from env), created_at.
strategies	Registered plugins: name, class_path, params_json, enabled.
bot_runs	One row per start→stop: started_at, stopped_at, environment, strategy, stop_reason (user error kill), started_by.
candles	Cached klines: symbol, interval, open_time PK, o/h/l/c/v.
trades	One row per round-trip: opened_at, closed_at, side (LONG SHORT), entry_px, exit_px, qty, fees, realized_pnl, r_multiple, exit_reason, strategy, environment, bot_run_id.
orders	Every exchange order: binance_order_id, trade_id, type, status, price, stop_price, qty, reduce_only, placed_at, filled_at, raw_json.
equity_snapshots	Every 4h close: ts, environment, balance, unrealized_pnl, month_to_date_pnl, sleeve_month_pnl.
events	Full audit: ts, level, category (trade bot breaker error sms news), message, payload_json, sms_status.
news_items / briefings	URL-unique raw items; daily Claude summaries with source links.

10. Notification service — notify.lk SMS

notify.lk HTTP API (user_id + api_key + sender_id, stored like exchange credentials). Fire-and-log: SMS failure never blocks trading; failures are retried 3× then logged with a dashboard banner.

Event	Example SMS
Trade opened	CryptoPilot: LONG opened 0.012 BTC @ 63,120. Stop 61,450, TP1 64,790. (DEMO)
Short opened	CryptoPilot: SHORT sleeve opened 0.008 BTC @ 63,120 (48% of equity, vol-scaled). No price stop; covers on regime end.
Trade closed	CryptoPilot: LONG closed +2.1% (+\$21.30), reason: trailing stop. Month: +4.2%.
Short resized	CryptoPilot: SHORT reduced to 31% of equity (vol spike).
Bot started	CryptoPilot: bot STARTED on LIVE, strategy trend_rider_v6, equity \$1,024.
Bot stopped	CryptoPilot: bot STOPPED by user. Open position left running with its exchange stops.
Breaker	CryptoPilot: monthly loss cap hit (−4%). All closed. Halted until the 1st.
Error	CryptoPilot ALERT: order rejected (−2019 margin). Bot paused — check dashboard.

11. AI news assistant

A fully separate module — shares only the database and scheduler; no access to trading code or API keys. Pipeline (daily at the configured local time, plus on-demand refresh):

- **Collect** — configured sources: crypto RSS (CoinDesk, CoinTelegraph, Bitcoin Magazine), macro headlines, plus the static economic calendar (FOMC decision dates, CPI release dates).
- **Deduplicate & store** — URL-unique insert into news_items.
- **Summarise** — the day's items go to the Claude API (claude-haiku-4-5 default; claude-sonnet-5 configurable) with a fixed prompt: 5–8 market-relevant bullets, neutral tone, flag anything affecting BTC volatility (regulation, ETF flows, Fed).
- **Publish** — briefing stored and shown on the dashboard with source links and the upcoming FOMC/CPI panel.

Design rule: the agent is read-only context for the human. Backtests showed event-based trading filters lose money (IMPROVEMENT_REPORT.md), so news never feeds the strategy. Its practical value on event days: the owner knows a violent candle is scheduled, so a stop wick-through is expected behaviour, not a malfunction.

12. Dashboard specification

Screen	Contents
Login	Email + password (argon2), JWT session, optional TOTP 2FA.
Overview	Environment badge (DEMO/LIVE), bot status + Start/Stop + kill-switch, account balance, open position card (side, entry, size, stop or 'size-managed' for shorts, TP1, unrealized P&L;), this-month P&L;,, both breaker meters, equity curve, today's briefing.
Trades	Filterable table (side, environment, strategy, month) with per-trade drawer showing orders/fills; CSV export.
Monthly	Calendar-month table: trades, realized P&L;,, fees, breaker status; manual mark-withdrawn action.
News	Daily briefing archive, upcoming FOMC/CPI panel, source links.
Events	Audit log with level/category filters.
Settings	Environment switch (bot must be stopped), API keys (write-only), strategy + parameters, risk %, sleeve weight/vol target, leverage cap, SMS toggles/templates, news sources/time.

Live updates via WebSocket (position, P&L;,, bot state) so the overview is real-time without refreshing.

13. Security design

- Binance API keys created with **trade + read only** — **withdrawals disabled**; IP-whitelisted to the VPS.
- Secrets encrypted at rest (AES-GCM) with a master key provided only via environment variable; keys are write-only in the UI (never displayed back).
- TLS everywhere (Caddy/Let's Encrypt); dashboard behind JWT + argon2 + optional TOTP; single-owner role.
- Kill-switch reachable in two clicks from any dashboard screen.
- Server hardening: non-root containers, firewall (only 443 + SSH), fail2ban, unattended security updates.
- Backups: nightly encrypted pg_dump to object storage; restore drill part of acceptance.

14. Testing and rollout plan

Phase	What happens	Exit criteria
1. Parity tests	The bot's strategy module replays the 3-year candle history offline; its decisions are diffed bar-by-bar against the validated research engine (final_composite.py).	Zero decision mismatches; all unit tests green.
2. Testnet dry-run	Full stack on Binance DEMO with small equity; all SMS live; forced-failure drills (network cut, restart mid-trade, kill-switch, breaker).	≥4 weeks; every trade explained; reconciliation clean; SMS ≤60s.
3. LIVE pilot	Switch to LIVE in Settings; minimum sizes (1–2% risk, sleeve weight 50%).	4+ weeks live matching DEMO behaviour; owner sign-off.
4. Full operation	Raise to validated defaults (3–5% risk, sleeve 75%) if desired.	Ongoing monthly review vs Appendix A expectations.

15. Deployment and operations

- Docker Compose: backend (FastAPI + bot + scheduler), postgres, caddy (TLS), dashboard (static build).
- VPS: 1 vCPU / 2 GB RAM (e.g. Hetzner/DO \$6–8/mo); region near Binance endpoints for stable latency.
- Monitoring: healthcheck endpoint + dead-man's-switch cron (missed 4h tick → SMS); container auto-restart.
- Logs: structured JSON to stdout, shipped to the events table for anything user-relevant.
- Upgrades: blue/green — never upgrade with an open position unless the change is hotfix-critical.

16. Project plan and estimates

#	Work package	Est. effort
WP1	Repo/CI, Docker skeleton, DB schema + migrations	3–4 d
WP2	Binance client (REST+WS, testnet/live), candle cache, reconciliation	4–5 d
WP3	Strategy framework + Trend Rider v6 plugin + parity tests	4–5 d
WP4	Risk & execution engine (sizing, vol targeting, breakers, orders)	5–6 d
WP5	FastAPI + auth + WebSocket + bot lifecycle	3–4 d
WP6	React dashboard (all screens)	6–8 d
WP7	Notifier (notify.lk) + templates + event log	1–2 d
WP8	AI news agent + briefing UI	2–3 d
WP9	Hardening, backups, failure drills, docs	3–4 d
	Total (single developer)	≈ 6–7 weeks

17. Operating costs

Item	Monthly estimate
VPS (1 vCPU / 2 GB)	\$6–10
notify.lk SMS (~60–120 msgs/mo)	LKR 300–700 (≈ \$1–2.5)
Claude API (daily haiku briefing)	\$1–3
Backup object storage	\$1
Binance fees	Included in strategy results (0.04–0.05%/side taker)
Total fixed	≈ \$10–17 / month

18. Risks and honest limitations

Risk	Mitigation / honest statement
Strategy stops working	Trend edges decay. Monthly review vs Appendix A; breakers cap damage; kill-switch always available. No guarantee of future returns.
Short sleeve has no price stop	By validated design (stops destroy the short edge). Risk is bounded instead by vol-targeted size ($\leq 75\%$ of equity, shrinks in crashes) + the -4% monthly sleeve breaker + regime exit. Worst backtested month -7.9% .
Funding rates not modeled	Shorts usually RECEIVE funding in bears (helps), but this is unmodeled; live funding is recorded per trade and reviewed monthly.
Exchange/technical outage	Reconciliation + safe-mode + SMS; exchange-resident stop orders protect the long book even if the VPS dies. The stop-free short is covered by the dead-man's-switch alert to the owner.
Wick-through / slippage	Stops are STOP_MARKET; backtest assumed conservative fills and survives $2.5\times$ fee/slippage stress.

Risk	Mitigation / honest statement
Key compromise	No withdrawal permission + IP whitelist bounds worst case to bad trades; kill-switch + key rotation runbook.
Regulatory/venue risk	Binance availability in the owner's jurisdiction is the owner's responsibility; testnet has no such exposure.
Owner expectation risk	23/37 backtested months green; single-digit red months WILL happen (worst -7.9%). 'Profit every month' does not exist honestly — documented three times in the research reports.

19. Future enhancements

- A/B parallel strategies on split equity (framework already supports registration).
- Funding-rate capture reporting; ETH sleeve (structure validated OOS-positive on ETH).
- Telegram bot as second notification channel; monthly PDF performance report.
- Withdrawal planner UI implementing the validated 10% monthly-profit rule (manual execution only).

Appendix A — Trend Rider v6 strategy (validated results)

Data & method: real Binance BTCUSDT 4h klines 2023-06 → 2026-07, fees 0.04–0.05%/side included, conservative fills, in-sample / out-of-sample discipline (OOS = the untouched 2025-07 → 2026-07 bear year). Full research: LONGSHORT_REPORT.md, BACKTEST_REPORT.md, IMPROVEMENT_REPORT.md.

Rules

- **Long engine (unchanged v5.2):** long only while close > SMA200 and EMA50 > EMA200 (4h). Entries: fresh regime or EMA20 pullback-resumption. Stop $2.5 \times \text{ATR}(14)$; TP1 at 1R sells 40% and stop moves to breakeven; runner trails at highest-high – $4 \times \text{ATR}$; exit all if the regime dies; –4% monthly breaker.
- **Short sleeve (v6):** short while close < SMA200 – $0.5 \times \text{ATR}$ and EMA50 < EMA200. Size = $75\% \times \min(1, 40\% / \text{realized vol})$ of equity (EWMA 48-bar annualized). NO price stop; resize on >20% drift; cover when the condition fails on a close; –4% monthly sleeve breaker. Long and short regimes are mutually exclusive — no leverage stacking.

Validated performance (fees included)

Metric	Composite v6	v5.2 long-only	Buy & hold
Total return (3y)	+204.5%	+137.3%	+134.6%
CAGR	+42.7%	+31.8%	+31.3%
Sharpe	1.48	1.45	0.82
Max drawdown	–17.2%	–13.2%	–53.4%
Worst month	–7.9%	–5.6%	–20.4%
Green months	23/37	16/37	21/37
OOS bear year 2025-07→2026-07	+31.7% (DD –13.8%)	–2.1%	–41.4%
Log-equity linearity R^2	0.91	0.82	0.55

Robustness evidence (why this is not curve-fit)

- Depth filter 0–1 ATR, vol target 0.3–0.5 (or off), sleeve weight 0.5–1.0: ALL profitable in both windows (full +174% to +224%, OOS +20% to +44%) — a plateau, not a spike.
- Survives 0.10%/side costs (2.5× actual). Same structure OOS-positive on ETH (+34%) and BNB (+51%) with BTC-tuned parameters untouched.
- Sleeve weight is the smoothness dial: 50% → Sharpe 1.54 / DD –15.9%; 75% (default) → balanced; 100% → +224% / OOS +44%. Halving overall size halves every drawdown and roughly halves returns.

Why the shorts have no stop-loss: every stopped-short variant tested loses — violent bear rallies whipsaw stops. The validated design controls short risk by position size (vol targeting shrinks the short automatically in crashes), the monthly sleeve cap, and the regime exit. Do not add a stop in implementation.